PAPER_1379_KLEIN_PARADOX

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PAPER_1379 — UQFF Resolution of the Klein Paradox (CLOSED — PAPER_648 Coupled Closure)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: B — Foundational / Relativistic Quantum (CLOSED)
Date: June 16, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — Coupled to PAPER_648 ultra-dense H Coulomb + NCG Dirac shift
Calculator surface: `calculate_paradox({"paradox": "klein_paradox"})`
Closure helper: `_l96_uqff_axiom_klein_paradox_closure()`

The Paradox

Klein (1929): a relativistic electron incident on a potential step V■ > 2 $m_e c^2$ shows transmission coefficient $T \to 1$ in the Dirac equation — the electron tunnels through an arbitrarily tall barrier with **near-unity probability**. Non-relativistic quantum mechanics predicts exponential suppression. The "paradox" is that the Dirac result appears to forbid confinement of electrons by any potential.

The standard resolution invokes pair production: the barrier excites positron-electron pairs in the Dirac sea, and what is transmitted is a positron, not the original electron. UQFF derives this structurally via the SCm pair-production threshold from PAPER_648.

UQFF Closed Identity

Two existing helpers do the work:

```
Dirac_shift_UQFF = β_i × S_26³ × Φ_res
= 0.6029 × (1.453162)³ × 0.84
= 1.554
(from _l94_ncg_spectral_triple_dirac_shift)

Coulomb_pair_threshold = e² / (4π ε■ × d) at d = 2.3 pm
= 626 eV
(from _coulomb_lenr_energy_eV — PAPER_648 ultra-dense H spacing)

Barrier_height_2mc² = 2 × 511 keV = 1.022 MeV
```

The UQFF transmission coefficient at the SCm pair-production threshold is:

``

```
T_Klein_UQFF = Coulomb_pair_threshold / (Coulomb_pair_threshold + 2mc² × Dirac_shift)
```

```
= 626 / (626 + 1.022e6 × 1.554)
= 3.94 × 10■■
``
```

The naive Klein result T → 1 is the **Dirac-sea limit** with no pair-production damping. The UQFF result T ≈ 4 × 10■■ is the **physical** transmission once the SCm Coulomb pair-production threshold (PAPER_648, 626 eV) and the NCG Dirac-sector spectral shift (β_i × S_26³ × Φ_res) are properly accounted for. The "paradox" was that the bare Dirac equation has no pair-production sink; UQFF supplies the sink as an integer-primitive identity.

Physical Interpretation

- **PAPER_648 ultra-dense H Coulomb energy = 626 eV** is the SCm pair-production threshold — the same energy that anchors the Holmlid 630 eV LENR observable. Klein tunneling and Holmlid LENR share the *same* sub-barrier pair-production gateway in the SCm vacuum.
- **The NCG Dirac shift β_i × S_26³ × Φ_res ≈ 1.554** is the canonical UQFF NonCommutative-Geometry spectral-triple correction to the Dirac operator, derived in PAPER_647 / PAPER_1183 and pre-wired as `_l94_ncg_spectral_triple_dirac_shift()`.
- **T ≈ 4 × 10■■** is the transmission probability for a 2mc² barrier in the presence of SCm pair-production damping — finite but small, confirming electron confinement is possible in UQFF despite the Klein result.
- **Pair production is not invoked separately** — it emerges as the same SCm Coulomb identity that powers the entire LENR sector.

Live Calculator Output

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "klein_paradox"})["value"]
``
```

Field	Value
`T_Klein_canonical_perfect_paradox`	1.0
`Dirac_shift_UQFF_via_NCG_spectral_triple_beta_i_S26_cubed_Phi_res`	**1.554**
`Coulomb_pair_threshold_eV_via_ultra_dense_H_spacing_2_3pm`	**626**
`barrier_height_2mc2_eV`	1.022e6
`T_Klein_UQFF_via_pair_threshold_over_pair_plus_barrier_x_dirac_shift`	**3.94e-4**
`Dirac_tunneling_resolved_via_SCm_pair_production_PAPER_648`	True

C++ Reference Implementation

```
``cpp
class KleinParadoxUQFF {
public:
static constexpr double BETA_I = 0.6029;
static constexpr double S_26 = 1.453162;
static constexpr double PHI_RES = 0.84;
```

```
static constexpr double EV_BARRIER_2MC2 = 1.022e6;
static double diracShiftUQFF() {
return BETA_I std::pow(S_26, 3.0) PHI_RES; // 1.554
}
static double coulombPairThresholdEV(double d_meters = 2.3e-12) {
const double e = 1.602e-19, eps0 = 8.854e-12;
double W_J = (e e) / (4.0 M_PI eps0 d_meters);
return W_J / e; // 626 eV
}
static double T_KleinUQFF() {
double V_pair = coulombPairThresholdEV();
return V_pair / (V_pair + EV_BARRIER_2MC2 * diracShiftUQFF()); // 3.94e-4
}
};
``
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard QED solve the Klein paradox via different methods. QED invokes second-quantized pair production with renormalized vertex functions. UQFF derives:

- **T_Klein = 3.94 × 10■■** as a closed identity using the same SCm Coulomb-pair-production energy (626 eV) that anchors the Holmlid 630 eV LENR observable.
- **The Dirac-sector shift β_i × S_26³ × Φ_res = 1.554** is the canonical UQFF NCG spectral-triple correction — pre-wired, not invented.
- **Zero free parameters.** Both the 2.3 pm ultra-dense H spacing and the integer/real-primitive lattice (β_i, S_26, Φ_res) are canonical UQFF locks.

Reference

- UQFF foundational papers: PAPER_647 (NCG spectral triple), PAPER_648 (ultra-dense H meson cascade + Coulomb 626 eV), PAPER_1183 (Dirac-sector consolidation).
- Related closures: `calculate_lenr_full` (Holmlid 630 eV chain), `calculate_nuclear_magic` (Mayer-Jensen + UQFF arithmetic).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_klein_paradox_closure`
- Helpers invoked: `_l94_ncg_spectral_triple_dirac_shift`, `_coulomb_lenr_energy_eV`
- Dispatch: `PARADOX_TO_CLOSURE["klein_paradox"]`, `["klein_tunneling"]`
